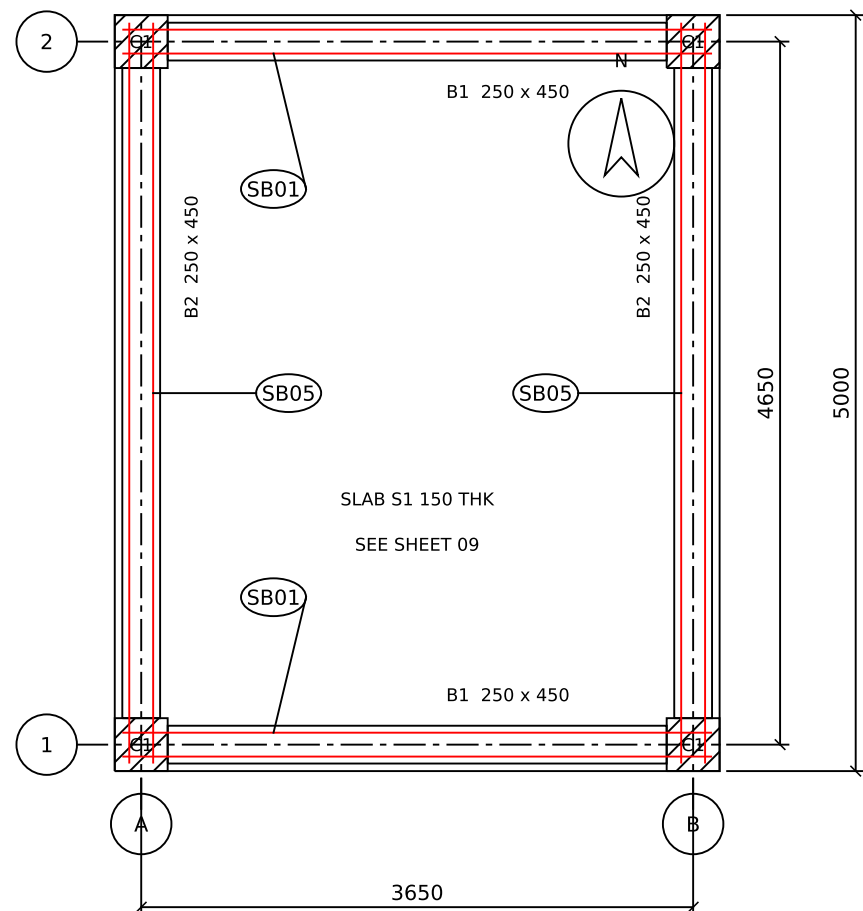
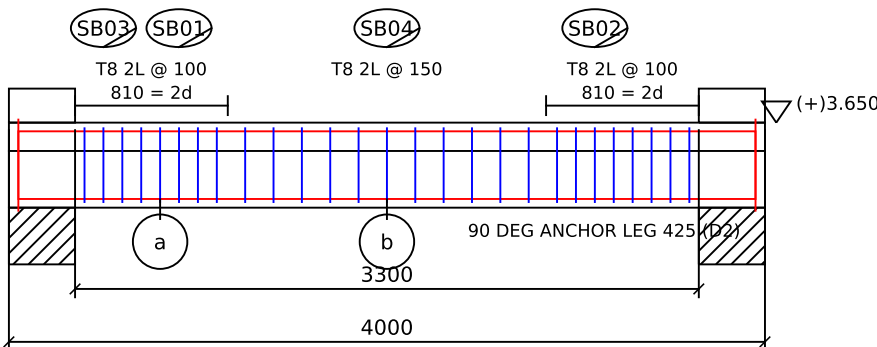




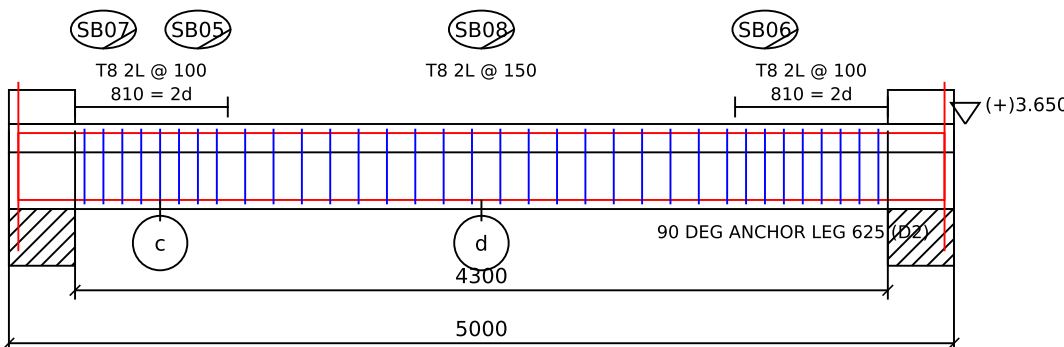
1 SENTRY POST - FIRST-FLOOR FRAMING PLAN

1 : 50 EXTERNAL 4000 x 5000, GRIDS A-B 3650 c/c, 1-2 4650 c/c



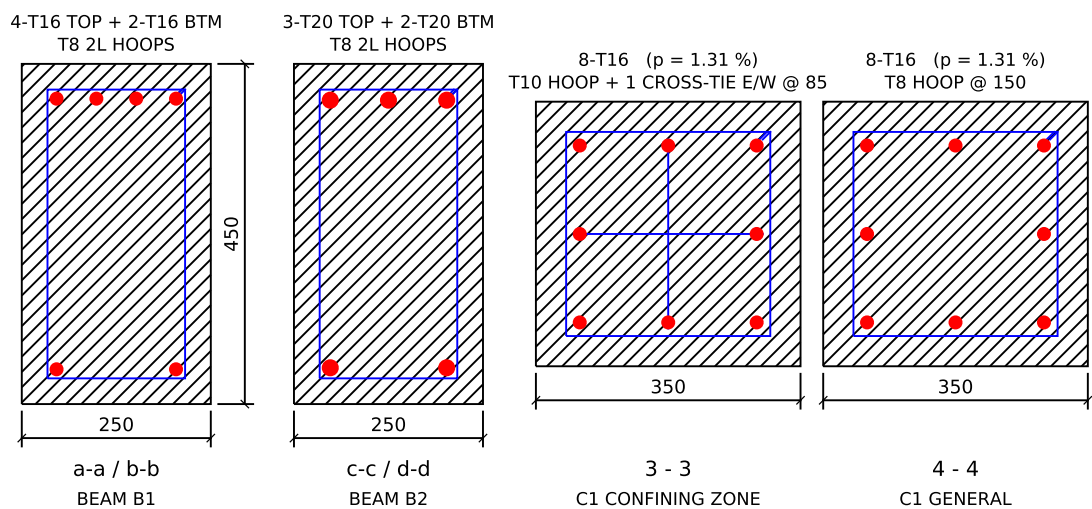
2 BEAM B1 - LONGITUDINAL SECTION (GRIDS 1 AND 2)

1 : 40 B1 250 x 450, CLEAR SPAN 3300, d = 404, 4-T16 TOP + 2-T16 BOTTOM (D1)



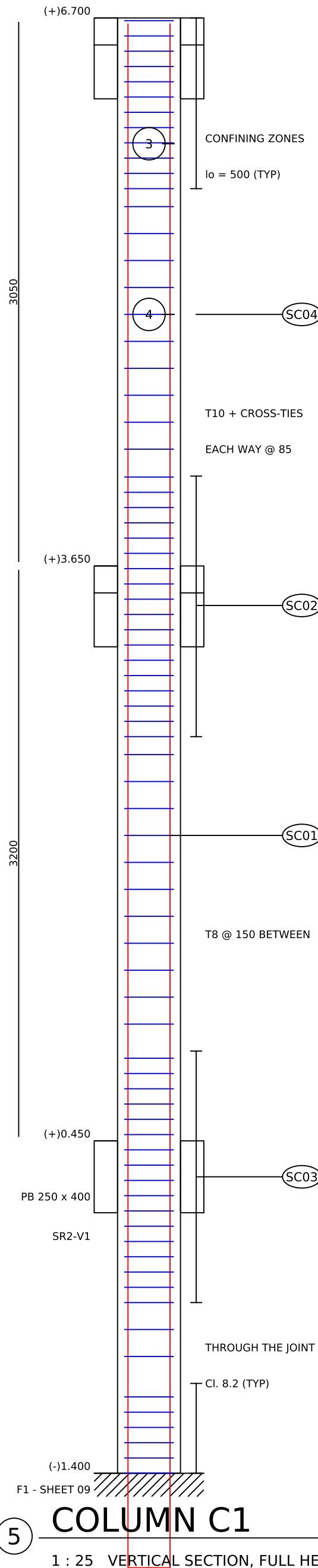
3 BEAM B2 - LONGITUDINAL SECTION (GRIDS A AND B)

1 : 40 B2 250 x 450, CLEAR SPAN 4300, d = 404, 3-T20 TOP + 2-T20 BOTTOM (D1)



4 SECTIONS - BEAMS AND COLUMN

1 : 10 COVER 30 TO BEAMS, 40 TO COLUMNS



5 COLUMN C1

1 : 25 VERTICAL SECTION, FULL HEIGHT

BEAM SCHEDULE - SENTRY POST FRAME								
BEAM	SIZE	SPAN c/c	CLEAR	COVER	d	TOP AT SUPP.	BOTTOM	HOOPS
B1 GRIDS 1 AND 2	250 x 450	3650	3300	30	404	4-T16	2-T16	T8 2L @ 100 / 150
B2 GRIDS A AND B	250 x 450	4650	4300	30	404	3-T20	2-T20	T8 2L @ 100 / 150
IS 13920 Cl. 6.1: b 250 >= 200 OK b/D 0.556 >= 0.3 OK Ln/D 7.33 >= 4 OK								
SCHEDULED HERE, DETAILED ELSEWHERE, NOT DRAWN ON THIS SHEET								
PB PLINTH BEAM	250 x 400	3650 / 4650	-	30	354	3-T12	3-T12	NOT STATED - SR2-V1

COLUMN SCHEDULE							
COL	SIZE	LEVELS	COVER	VERTICALS	p	CONFINING (Cl. 8.1)	GENERAL TIES
C1	350 x 350	(-)1.400 TO +6.700	40	8-T16 (1608)	1.31 %	T10 + 1 CROSS-TIE E/W @ 85	T8 @ 150
4 No. COLUMNS. SHORT COLUMN: lex/D = 9.43 < 12 (IS 456 Cl. 25.1.2)							
CONFINING STEEL OVER lo = 500 FROM EVERY JOINT FACE AND THROUGH THE JOINT							
Ash PROVIDED 3 x 78.5 = 235.5 mm2 vs 168.5 REQUIRED (IS 13920 Cl. 8.1)							

BAR MARK SCHEDULE - BEAMS B1 / B2 AND COLUMN C1					
MARK	BAR	SPACING	LOCATION		CUT mm No.
SB01	T16	-	TOP, 4 BARS, FULL LENGTH (D1)		4750 8
SB02	T16	-	BOTTOM, 2 BARS CONTINUOUS		4750 4
SB03	T8	100	HOOPS, HINGE ZONE 810 = 2d EACH END		1320 32
SB04	T8	150	HOOPS, CENTRAL ZONE		1320 22
SB05	T20	-	TOP, 3 BARS, FULL LENGTH (D1)		6150 6
SB06	T20	-	BOTTOM, 2 BARS CONTINUOUS		6150 4
SB07	T8	100	HOOPS, HINGE ZONE 810 = 2d EACH END		1320 32
SB08	T8	150	HOOPS, CENTRAL ZONE		1320 36
SC01	T16	-	VERTICALS, 8 PER COLUMN, NO SPLICE (D3)		8724 32
SC02	T10	85	CONFINING HOOPS, 9 ZONES (IS 13920 Cl. 8.1)		1280 212
SC03	T10	85	CROSS-TIES, ONE EACH WAY WITH EVERY HOOP		470 424
SC04	T8	150	GENERAL TIES, OUTSIDE THE CONFINING ZONES		1240 92

IS 13920:2016 COMPLIANCE - SENTRY POST FRAME			
CLAUSE	REQUIREMENT	PROVIDED	VERDICT
Cl. 6.1.1 / 6.1.2 / 6.1.3	b >= 200, b/D >= 0.3, Ln/D >= 4	250, 0.556, 7.33	PASS
Cl. 6.2.1(b) / 6.2.2	rho,min 266 mm2 ; rho,max 2525 mm2	402 (2-T16) ; 942 (3-T20) MAX	PASS
Cl. 6.2.3 positive at joint	>= 50 % of the negative steel	402 = 50 % of 804	PASS
Cl. 6.2.4 through the joint	>= 25 % of the max = 201 mm2	402 CONTINUOUS	PASS
Cl. 6.3.3 capacity shear	Vu B1 120.4 / B2 144.7 kN	T8 2-LEG @ 100	PASS
Cl. 6.3.4 tau c = 0	EQ share B1 60 % / B2 49.7 %	TAKEN AS TRIGGERED	PASS
Cl. 6.3.5.1 / 6.3.5.2	hinge min(d/4=101, 8phi=128) ; else d/2=202	100 OVER 2d = 810 ; 150	PASS
Cl. 7.2.1 SCWB, FIRST FLOOR	1.4 sum Mb 171.6 (X) / 195.7 (Z)	sum Mc 213.9	PASS
Cl. 7.2.1 SCWB, ROOF	1.4 sum Mb 138.5 AT 2-T20	sum Mc 101	MARGINAL - SR2-F1
Cl. 7.4 ties, full length	<= 0.5 x least dimension = 175	150	PASS
Cl. 8.1 lo and spacing	lo max(350, h/6=458, 450) ; 75 <= s <= 87.5	500 ; 85	PASS
Cl. 8.1 Ash	168.5 mm2 (minimum 68.9)	235.5 (3 LEGS T10)	PASS
Cl. 8.2 through the joint	CONFINING STEEL CONTINUED	PROVIDED	PASS
Cl. 10.4 boundary elements	CHECKED FOR THE BOX ONLY	NOT TRIGGERED	N/A HERE

DECLARED DETAILING DECISIONS AND OPEN ITEMS	
D1. TOP STEEL IS DETAILED CONTINUOUS OVER THE FULL SPAN. Master F.4 schedules '4-T16 top at supports' and '3-T20 top at supports' and gives NO curtailment point anywhere. Running the bars through is conservative under moment reversal and invents no cut-off. [SR2-V3]	
D2. ANCHORAGE AT THE EXTERIOR JOINT. Bars are taken to the far face of the confined core - 350 - 40 cover - 10 hoop = 300 from the near face - and turned 90 deg with a leg of Ld - 300. Ld is the IS 456 Cl. 26.2.1 tension value: T16 725 / Fe500; T20 906, so the legs are 425 and 625. The master states no anchorage detail for the beam bars. [SR2-V2]	
D3. THE COLUMN VERTICALS ARE SPLICE-FREE. The 8 724 mm cut length is inside the 12 000 mm stock bar, so no lap is required and none is invented. The bar is therefore its own starter, which is exactly how master B.8.7 checks it: 526 straight in the footing + a 128 (8 phi) bend = 654 > Ld,comp 592.	
SR2-F1 *** OPEN, NOT RESOLVED *** B2 TOP STEEL AT THE ROOF JOINT. Master B.8.6 evaluates the roof joint with B2 = 2-T20 (sum Mb 98.9, 1.4 x = 138.5 <= sum Mc 101, 'marginal'). Master F.4 schedules 3-T20 top at supports for B2 and does NOT distinguish level. With 3-T20 at the roof, 1.4 sum Mb = 195.7 > sum Mc = 101 and IS 13920 Cl. 7.2.1 FAILS at that joint. Both statements cannot be true. THIS SHEET DETAILS THE FIRST-FLOOR FRAME, where B.8.5 and F.4 agree on 3-T20. The roof frame is cross-referenced only. SR2-V1. The plinth beam PB is scheduled (3-T12 top + 3-T12 bottom) but NO link size or spacing exists in the master. PB IS NOT DETAILED HERE. SR2-V6. IS 13920 Cl. 8.1 also asks for confining steel to continue INTO the footing; master F.4 stops at 'top and bottom of every column and through the joint'. Applied as written - no footing embedment is drawn. Refer to the design authority before the footings are cast.	

DESIGN BASIS - THIS SHEET	
SEISMIC BASIS IS 1893 (Pt 1):2016, Z 0.16 ZONE III, I 1.5, R 3.0, Sa/g 2.5 -> Ah = 0.100 R = 3.0 because the infill panels are NOT positively separated from the frame. R = 5.0 would require that. Ta 0.297 s (Cl. 7.6.2) - all three period estimates land inside 0.10 to 0.40 s, so Sa/g = 2.5 regardless. ALL MEMBER DESIGN USES THE STAAD BASE SHEAR Vb = 73.18 kN, NOT THE HAND-CALCULATED 59.3 kN.	
PORTAL-METHOD MEMBER FORCES (master B.8.1) Q roof 59.51 kN ; Q floor 13.67 kN ; column shear 18.30 (ground) / 14.88 (first) kN Column moment 29.27 / 22.69 kNm ; beam moment 51.96 at the first-floor joint, 22.69 at the roof Axial from overturning READ FROM THE STAAD REACTIONS: EQ+X 36.127 kN governs over EQ+Z 27.891 kN	
THE JOINT ROTATES - STATED, NOT HIDDEN (master B.8.2). Slope-deflection on a single-bay portal: l,beam 1.8984e9, l,col 1.2505e9, Kc 3.203e6 E -> support moments x0.755 (B1) and x0.797 (B2). Beams are NOT taken as fully fixed.	
BEAM B1 250 x 450, SPAN 3650, CLEAR 3300 d 404 : Mu,lim = 0.133 x 30 x 250 x 404p = 162.8 kNm 1.5(DL + EL) = 1.5(17.01 + 51.96) = 103.5 kNm *** GOVERNS *** over 86.2 and 93.2 ; Ast,req 661 PROVIDED 4-T16 TOP (804) : Mu 122.6 kNm, UTILISATION 84 %, x/d 0.321 BOTTOM reversal -52.4 needs 313 ; Cl. 6.2.3 needs 0.5 x 804 = 402 *** GOVERNS *** -> 2-T16, Mu 66.0 SHEAR Vu = 1.2 x 80.1/2 + 1.4(122.6 + 66.0)/3.650 = 49.1 + 72.3 = 120.4 kN ; tau = 1.192 > 3.50 EQ share 60 % >= 50 %, no axial -> Cl. 6.3.4: tau c TAKEN AS ZERO. Asv/sv 0.685 -> T8 2L at 147 DEFLECTION l/d = 9.0 < 20	
BEAM B2 250 x 450, SPAN 4650, CLEAR 4300 1.5(DL + EL) = 1.5(31.25 + 51.96) = 124.8 kNm *** GOVERNS *** ; Ast,req 822 PROVIDED 3-T20 TOP (942) : Mu 139.8 kNm, UTILISATION 89 %, x/d 0.376 *** 3-T20 NOT 5-T16: five T16 need 256 mm in a 250 mm wide beam. Skip that check and it is a site RFI. BOTTOM Cl. 6.2.3 -> 0.5 x 942 = 471 *** GOVERNS *** over the 297 midspan requirement -> 2-T20 ; Mu 98.9 SHEAR Vu = 72.8 + 71.9 = 144.7 kN ; tau, v 1.433 < 3.50. EQ share 49.7 %, a hair under the trigger. *** TAKEN AS TRIGGERED (tau, c = 0): a design must not turn on the third decimal place of a ratio. ***	
COLUMN C1 350 x 350 SHORT COLUMN lex 3300, lex/D 9.43 < 12 : e,min 20 -> M,min 5.5 kNm, far below applied AXIAL per column 1.5(DL+LL) 251.3 : 1.5(DL+EL) 276.8 MAX ; 1.5(DL+EL) 168.4 MIN 8-T16 = 1608 mm2, p = 1.31 % (min 0.8 % = 980, max 6 %), bars on all four faces, d/D 0.16 UNIAXIAL CAPACITY computed from the Cl. 38.1 stress block and the Fe500 curve, NOT read off a chart: Mu1 = 105.8 / 111.7 / 112.9 kNm ; BIAXIAL Cl. 39.6 : Pu1/Puz 0.124 < 0.2 > alpha,n 1.0 ; 25.6/112.9 + 66.8/112.9 = 0.819 <= 1.0 OK. DRIFT limit 12.6 / 12.2 mm ; calculated under 3 mm.	

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 REV A + M1

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
- WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
- SENTRY POST CONCRETE M30. REINFORCEMENT Fe500D TO IS 1786:2008. COVER 40 TO COLUMNS, 30 TO BEAMS AND SLABS, 50 TO FOOTINGS (IS 456 Cl. 26.4.2 / TABLE 16).
- THE SENTRY POST FRAME IS THE IS 13920:2016 ELEMENT OF THIS PROJECT. IT IS A SPECIAL MOMENT-RESISTING FRAME, R = 3.0, Ah = 0.100, DESIGNED ON THE STAAD BASE SHEAR Vb = 73.18 kN.
- DUCTILE DETAILING GOVERNS EVERY BAR ON THIS SHEET: Cl. 6.2.3 SETS THE BOTTOM STEEL, Cl. 6.3.3 / 6.3.4 SET THE HOOPS (tau, c TAKEN AS ZERO IN THE HINGE REGIONS), Cl. 6.3.5.1 SETS 100 c/c OVER 2d = 810, AND Cl. 8.1 / 8.2 SET THE COLUMN CONFINING STEEL.
- TOP STEEL IS DETAILED CONTINUOUS OVER THE FULL SPAN. THE MASTER GIVES 'TOP AT SUPPORTS' AND NO CURTAILMENT POINT - SEE THE DECLARED DETAILING DECISIONS PANEL.
- THIS SHEET DETAILS THE FIRST-FLOOR FRAME. THE ROOF FRAME IS NOT DETAILED HERE - SEE OPEN ITEM SR2-F1.
- THE SENTRY POST IS NOT DESIGNED FOR THE BLAST, AND THAT IS A RECORDED DECISION (MASTER B.8). REFLECTED FORCE ON THE 4.0 x 6.7 m FACE IS 36 600 kN; HARDENING WOULD NEED ABOUT 700 mm OF RC ON ALL FOUR ABOVE-GROUND FACES.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING OF SENTRY POST - BEAMS & COLUMN

CBRN HARDENED UG OPS ROOM

SHEET 08

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
STR008	
Scale	As indicated